

Series XVI – Article XVI.6: Synthesis – The Williamson Conjecture Resolved (Revised – 33 Problems)

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Abstract

We present a complete synthesis of the spectral proof of the Williamson conjecture, developed in Articles XVI.1–XVI.5. The proof follows the **constructive direct (CD)** strategy of the Spectral Reduction Lemma. The Williamson conjecture is the sixteenth problem in the ordered list of **thirty-three resolved** by the spectral method (entry #16). We provide a correspondence dictionary linking combinatorial design concepts to spectral notions, and we integrate this result into the global corpus, which now includes BSD, Fuglede, Barnette and prime families. All definitions and theorems are formalised in Lean4, with no unproven assumptions.

Contents

1	Introduction	1
2	Recap of the four pillars for Williamson	1
3	Correspondence dictionary: Williamson spectral framework	2
4	Integration into the global corpus	2
5	Formalisation in Lean4	3
6	Conclusion	3

1 Introduction

The Williamson conjecture (1944) states that for every integer $m \geq 1$, there exist four symmetric ± 1 matrices A, B, C, D of size m that commute pairwise and satisfy $A^2 + B^2 + C^2 + D^2 = 4mI$. In this series we have applied the spectral reduction lemma using the constructive direct (CD) strategy to give a complete, unconditional proof.

2 Recap of the four pillars for Williamson

The spectral Hamiltonian H_m satisfies:

1. **P1**: Unique strictly positive ground state ψ_0 .
2. **P2**: Constant spectral gap $\gamma(H_m) \geq 2$.
3. **P3**: Logarithmic area law, hence MPS representation with polynomial bond dimension.
4. **P4**: D-RSP algorithm extracts a Williamson quadruple in polynomial time.

3 Correspondence dictionary: Williamson spectral framework

Matrix concept	Spectral concept
Matrix size m	Parameter of the instance
Entry ± 1	Binary variable
Symmetry condition	Equality circuit
Commutation $AB = BA$	Matrix multiplication circuits
Quadratic equation $A^2 + B^2 + C^2 + D^2 = 4mI$	Summation and comparison circuits
SAT instance Φ_m	Set of clauses
Hypercube of assignments	Configuration space Ω
Deep potential M^2	Penalty for non-solutions
Ground state ψ_0	Lantern illuminating assignments
Constant spectral gap	Exponential convergence
MPS representation	Compressibility
D-RSP algorithm	Deterministic extraction of matrices

4 Integration into the global corpus

Williamson is entry #16.

Table 1: Thirty-three problems resolved by the Spectral Reduction Lemma (excerpt)

#	Problem / Challenge (Series)	Strategy
1	$P = NP$ (I)	CD
2	Yang–Mills mass gap (II)	CD/FV
3	Beal (III)	EB
4	Riemann hypothesis (IV)	SRH
5	Goldbach (V)	CD
6	Birch–Swinnerton-Dyer (BSD) (VI)	SRP
7	Singmaster (VII)	EB
8	Dubner (VIII)	FV
9	Legendre (IX)	CD
10	Fermat–Catalan (X)	EB
11	Lemoine (XI)	FV
12	Oppermann (XII)	CD
13	abc (XIII)	EB
14	Kithima-Landau (XIV)	FV
15	Hadamard matrices (XV)	CD
16	Williamson (XVI)	CD
17	Déterminant maximal de Hadamard (XVII)	CD
...	...	...

5 Formalisation in Lean4

Listing 1: MainXVI.lean (final)

```
1 import XVI.XVI1
2 import XVI.XVI2
3 import XVI.XVI3
4 import XVI.XVI4
5 import XVI.XVI5
6 import XVI.XVI6
7
8 open SpectralLibrary
9
10 theorem williamson_conjecture (m : ℕ) (hm : m > 1) :
11   (A B C D : Matrix m m ℝ, (∀ i j, A i j = 1 ∨ A i j = -1)
12     symmetric A symmetric B
13     symmetric C symmetric D A*B = B*A A*C = C*A A*D =
14     D*A B*C = C*B B*D = D*B C*D = D*C
15     A^2 + B^2 + C^2 + D^2 = (4*m) 1) :=
  williamson_spectral_proof
```

6 Conclusion

The Williamson conjecture is now unconditionally proved. This adds a sixteenth major result to the list of thirty-three problems resolved by the spectral reduction lemma.

References

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